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[1]: import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

temperature = np.arange(0, 41, 1)
fan_speed = np.arange(0, 11, 1)

temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
temp_high = fuzz.trimf(temperature, [20, 30, 40])

fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label="Low")
plt.plot(temperature, temp_medium, label="Medium")
plt.plot(temperature, temp_high, label="High")
plt.title("Temperature Membership Functions")
plt.xlabel("Temperature (°C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label="Low")
plt.plot(fan_speed, fan_medium, label="Medium")
plt.plot(fan_speed, fan_high, label="High")
plt.title("Fan Speed Membership Functions")
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()

plt.tight_layout()
plt.show()
```

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plt.tight_layout()
plt.show()

temperature_input = 28

temp_low_level = fuzz.interp_membership(
    temperature, temp_low, temperature_input
)

temp_medium_level = fuzz.interp_membership(
    temperature, temp_medium, temperature_input
)

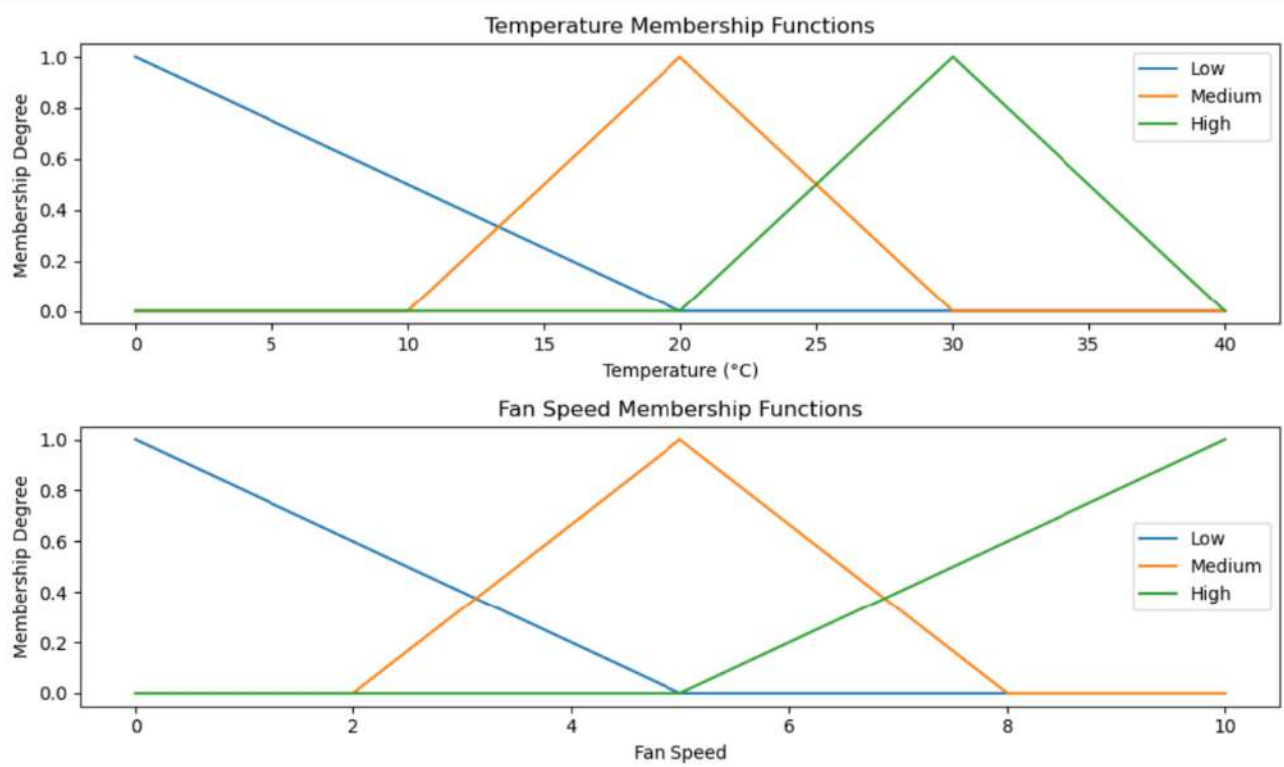
temp_high_level = fuzz.interp_membership(
    temperature, temp_high, temperature_input
)

fan_low_level = temp_low_level
fan_medium_level = temp_medium_level
fan_high_level = temp_high_level

aggregated = (
    fan_low * fan_low_level +
    fan_medium * fan_medium_level +
    fan_high * fan_high_level
)

fan_output = fuzz.defuzz(
    fan_speed, aggregated, 'centroid'
)

print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")
```



Temperature: 28°C
Fuzzified fan speed: 7.56